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Filling performance of an Acid Blue 1 leveler on blind microvias†

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The void-free blind microvia copper electroplating process is a key technology in HDI PCB manufacturing. This work systematically studied the influence of the Acid Blue 1 (AB1) leveler on blind microvia filling. AB1 exhibits excellent suppression during electroplating, with its effect strengthening as the AB1 concentration increases. Chronopotentiometry was used to study the interaction among AB1, SPS, and PEG. There is a synergistic suppressing interaction between the new leveler AB1 and suppressor PEG, while there is competition between AB1 and accelerator SPS. During electrodeposition, AB1 can adsorb on the copper substrate to reduce active sites and suppress copper electrodeposition, based on theoretical studies and *in situ* infrared spectroscopy. At the optimized electroplating process, void-free microvia filling is achieved with 20 mg L^{-1} AB1 and successive electroplating confirms the stability of AB1. Based on diffusion differences among additives and the competition between AB1 and SPS, a gradient competitive adsorption superfilling mechanism was proposed to illustrate the filling process of the AB1–SPS–PEG system. AB1 is beneficial to improving flatness and lowering surface roughness of the copper layer, which is conducive to reducing signal losses during high-frequency transmission. Additionally, AB1 can modulate the preferential formation of Cu(111) and Cu(200) and facilitate grain refinement.

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1. Introduction

High-density interconnect (HDI) printed circuit boards (PCBs) serve as a foundational platform for connecting all kinds of electronic components in electronic products, thus attracting significant research attention.[1] With the trend toward smaller-sized electronic devices with higher performance, stricter requirements are imposed on PCB design and manufacturing technology.[2,3] Especially in the fields of high-performance computing, high-speed communication, and precision instruments, the reliability and performance of PCBs are directly related to the performance and stability of the final products. Blind microvia filling is a key interconnect technology in the manufacturing process of PCBs, allowing for electrical connections between different layers in the circuit board.[4,5] The application of this technology enhances PCB design flexibility, enabling more complex circuit layouts and greater space utilization.

Copper is widely used as an interconnect line in blind microvias for PCB manufacturing due to its good electronic (thermal and electrical conductivity) and mechanical (wear resistance and elongation, *etc.*) properties.[6,7] The filling of blind microvias with copper is achieved through electroplating technology.[8–10] In PCBs, copper not only provides a stable electrical connection but also enhances the structural strength of HDI PCBs, improving heat and wear resistance.[11] Moreover, copper interconnect lines effectively reduce losses in signal transmission and enhance signal integrity, which is especially important for high-frequency and high-speed circuits.[12] During copper electroplating, it is essential to adjust the metal deposition process in local microvias to avoid the formation of void defects caused by the edge effect, which can undermine the properties of the copper interconnect layer. The most common method is the addition of proper additives.[13–15]

The additives used in copper electroplating fall into three categories: accelerators, suppressors, and levelers.[16] The most commonly used accelerator, suppressor, and leveler is bis-(sodium sulfopropyl)- disulfide (SPS), polyethylene glycol (PEG), and Janus Green B (JGB).[17] During electroplating, density functional theory (DFT) calculations, *in situ* Raman spectroscopy, and scanning tunneling microscopy (STM) studies reveal

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that the disulfide bond of SPS will break up, with the sulfur-containing side adsorbing on the copper base, while the other side (containing a sulfonic acid group) captures free Cu^{2+} to accelerate reduction with adsorbed Cl^- through the "bridging effect".[18–20] As for PEG, *in situ* Raman and molecular dynamics (MD) simulations confirm the formation of PEG–Cu^{2+}–Cl^- adsorbate complexes, which increase the energy barrier and suppress copper electrodeposition.[21–23] Mass spectrometry (MS) and DFT calculations demonstrate that the azo group of JGB competes with the reduction of Cu^{2+} to obtain electrons, leading to dissociation[24,25] into phenazine derivatives that can also adsorb on active sites to suppress copper growth.[26,27] Among these additives, levelers are key components for achieving defect-free filling in blind vias.[28] Nowadays, triphenylmethane derivatives are promising levelers and have been used in through-hole and blind-hole filling due to the unlocalized orbital distribution of the tertiary amine structure.[29,30]

In this work, a novel triphenylmethane derivative, Acid Blue 1 (AB1), was selected as a potential leveler, and the effects of different concentrations of AB1 on copper electroplating were investigated. The interactions among three different types of additives were studied using galvanostatic measurements. The influencing mechanism of AB1 on the copper cathode surface during electroplating was also studied theoretically and experimentally. The filling performance of AB1 in blind microvias was examined, and the electroplating process was optimized. Additionally, the surface morphology and roughness of the copper interconnect layer were analyzed, and finite element simulation was employed to study the high-frequency signal transmission losses influenced by AB1, while phase structure was investigated by XRD.

2. Experiment

2.1 Chemical materials

The chemicals used included $CuSO_4{\cdot}5H_2O$, H_2SO_4, and NaCl, which were purchased from Sinopharm Group. The virgin make-up solution (VMS) was composed of 250 g L^{-1} $CuSO_4{\cdot}5H_2O$, 40 g L^{-1} H_2SO_4, 60 mg L^{-1} Cl^-. The additives used in this work include Acid Blue 1 (AB1), PEG, and SPS, which were sourced from Shanghai Aladdin Biochemical Technology. All chemicals were analytical-grade, and all solutions prepared used deionized water.

2.2 Electrochemical tests

A traditional three-electrode system, consisting of a Pt rotating disk electrode (working electrode), Pt foil (counter electrode), and saturated mercury sulfate electrode (referenced as SSE; reference electrode), was used to measure cyclic voltammetry (CV), electrochemical impedance spectra (EIS), and galvanostatic measurements. All electrochemical tests were controlled by the Gamry Framework Station. The voltage range for CV was −0.8 V to 0.2 V *vs.* SSE, with a scan rate and rotation speed of 50 mV s^{-1} and 1000 rpm.[31] For the EIS tests, pre-deposition was performed with the WE deposited at 2 A dm^{-2} for 1 min in VMS. EIS was subsequently tested with the WE potential set at −0.5 V *vs.* SSE. The frequency range was from 10^5 Hz to 10^{-2} Hz, and the rotation speed was 1000 rpm.[32] Galvanostatic measurements were carried out at 2 A dm^{-2} and 1000 rpm with additives introduced into the solution at 500 s, 1000 s, and 1500 s in different injection orders.[33]

2.3 Theoretical calculations

Gaussian was used to optimize the AB1 structure with B3LYP at 6-311G,[34] while Multiwfn was employed to calculate the local electron affinity (LEA) to predict the adsorption area,[35,36] and the LEA distribution was visualized using VMD 1.9.3.[37] The theoretical calculation of AB1 adsorption on the copper substrate was performed using the Vienna *ab initio* simulation package (VASP).[38,39] A cubic cell measuring 23 Å in length and width, and 46 Å in height, was used for the calculation. The exchange–correlation function for structural relaxation was addressed using the GGA parameterized by PBE. The ion-electron interactions were determined through the PAW pseudopotential method, with the self-consistent field (SCF) convergence set to 10^{-5} eV.[40] The adsorption energy of the novel leveler AB1 on the copper substrate was calculated as follows:[41]

$$\Delta E = E_{AB1/Cu} - E_{Cu} - E_{AB1} \quad (1)$$

where $E_{AB1/Cu}$ is the total energy after AB1 adsorbs on the copper substrate, E_{Cu} is the energy of the copper substrate, and E_{AB1} is the energy of AB1. The molecular dynamics simulation for calculating the diffusion properties of AB1 is detailed in the ESI.†

2.4 Electroplating

In electroplating experiments, phosphor copper plates (5 × 10 cm^2) were used as the anode, while PCB plates (5 × 10 cm^2) served as the cathode. Gas pumping, with a rate of 20 L h^{-1}, was employed to stir the solution and accelerate mass transfer during electroplating. Each electroplating used 1.5 L solution, with the solution composition and optimized values shown in Table 1. Before electroplating, the PCB was cleaned with acetone and activated using 10 wt% H_2SO_4. The electroplating time for filling microvias was 3 h.

2.5 Copper characterization

In situ infrared spectroscopy was measured using a Nicolet IS50 IR with an MCT detector. The testing solution comprised 8 mmol L^{-1} $CuSO_4{\cdot}5H_2O$, 70 mmol L^{-1} H_2SO_4, and 1.4 mmol L^{-1} Cl^-. The potential was controlled by the Gamry Framework station and ranged from −0.45 V to −0.65 V *vs.* SSE.

Table 1 Electroplating solution composition and process

Solution composition and process	Optimized values
$CuSO_4{\cdot}5H_2O$ (g L^{-1})	220
H_2SO_4 (g L^{-1})	50
Cl^- (mg L^{-1})	60
Acid Blue 1 (mg L^{-1})	20
SPS (mg L^{-1})	2
PEG (mg L^{-1})	200
Cathode current density (A dm^{-2})	1.5

SEM (Zeiss Supra 55) was used to examine morphology, and an OLYMPUS DSX510 conformal microscope was used to measure roughness. The phase structure was studied by XRD (Brucker D8). The copper layer grain size was calculated based on Scherrer's formula:

$$D = \frac{K\lambda}{\beta \cos\theta} \quad (2)$$

where D is the grain size (nm), K is 0.89, β is the full width at half maxima in 2θ degree, and θ is the diffraction angle.

2.6 Finite element simulation

COMSOL Multiphysics software was used to carry out the finite element simulation. The model was established based on the surface roughness measurement. The AC/DC module was employed to simulate the current density distribution at varying high frequencies (10 GHz and 100 GHz).

3. Results and discussion

3.1 Electrochemical analysis

3.1.1 CV analysis. Cyclic voltammetry was used to examine AB1 during electroplating,[42] as shown in Fig. 1a and Fig. S1 (ESI†). The cathodic onset potential decreases as AB1 concentration increases (Fig. S1, ESI†). The onset potential without AB1 was about −430 mV, while it shifts negatively to −575 mV after adding 10 mg L^{-1} AB1, indicating that AB1 effectively increases cathodic polarization. In addition, the onset potential decreases from −583 mV to −650 mV with additive concentrations rising from 20 mg L^{-1} to 150 mg L^{-1}, demonstrating an intensified suppressing effect on copper electrodeposition with higher AB1 concentrations. The anodic stripping area also declines continuously with increasing AB1 concentration, suggesting a reduction in the amount of electrodeposited Cu during the cathodic process. We calculated the anodic stripping coulomb (ASC) values according to eqn (S1) for quantitative evaluation of AB1 (details in ESI†).[43] Fig. 1b shows that the ASC of VMS is 71.0 C dm^{-2}, which decreases by 28% to 51.4 C dm^{-2} with 10 mg L^{-1} AB1, indicating that trace AB1 (10 mg L^{-1}) can significantly suppress copper electrodeposition. At 20 mg L^{-1} and 30 mg L^{-1} AB1 concentrations, the ASC is 47.8 C dm^{-2} and 42.7 C dm^{-2}, both lower than 50 C dm^{-2}. The ASC further decreases to 35.8 C dm^{-2} with AB1 concentration at 50 mg L^{-1}. Notably, the ASC drops to 25.8 C dm^{-2}, accounting for 36% of VMS. Overall, cyclic voltammetry analysis indicates that AB1 suppresses copper electrodeposition.

3.1.2 EIS analysis. In addition to CV tests, electrochemical impedance spectra (EIS) were conducted to study the effect of AB1.[44] The impedance semicircle is related to charge transfer impedance (R_{ct}). R_{ct} increases with increasing impedance semicircle.

Fig. 1b (right) shows the influence of AB1 during EIS tests. The semicircle enlarges as AB1 concentration rises, indicating an increase in charge transfer impedance and a higher reaction transition energy requirement for the electroreduction of Cu^{2+} to Cu. Solution impedance (R_s) and charge transfer impedance (R_{ct}) were simulated according to the equivalent circuit shown in the inserted figure in Fig. 1b. The simulated results are depicted in Fig. 1b (left). Firstly, the generally unchanged R_s

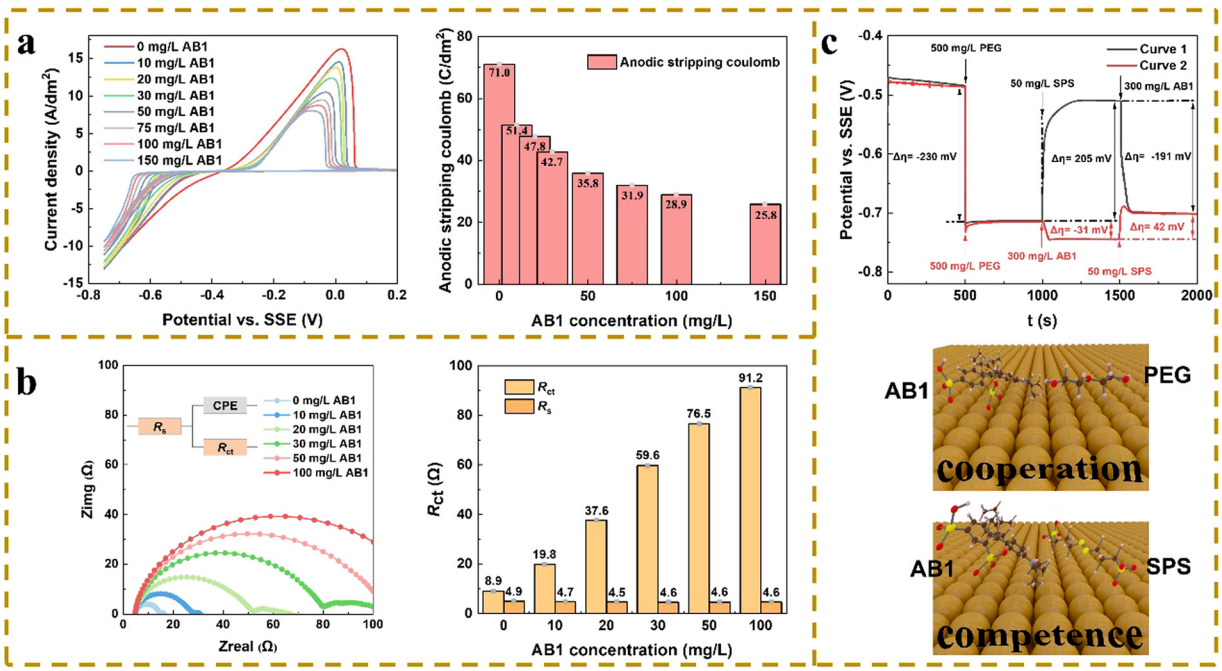


Fig. 1 (a) Cyclic voltammetry under the influence of AB1 leveler (left) and histogram of anodic stripping coulomb (right). (b) EIS of AB1 measured at −0.5 V *vs.* SSE and 1000 rpm, with the simulated equivalent circuit shown in the inserted figure (left) and histogram of simulated solution impedance (R_s) and charge transfer impedance (R_{ct}) (right). (c) Galvanostatic measurement with different additive injection orders at 2 A dm^{-2} and 1000 rpm (top) and schematic diagram of interactions among SPS, PEG, and AB1 (bottom).

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suggests that AB1 has little influence on solution conductivity. For R_{ct}, it is only 8.9 Ω for VMS, indicating that Cu^{2+} is easily electro-reduced during electroplating without AB1. Introducing 10 mg L^{-1} increases R_{ct} to 19.8 Ω, two times higher than VMS. R_{ct} continues to rise from 37.6 Ω to 91.2 Ω with AB1 concentrations of 20 mg L^{-1} to 100 mg L^{-1}, and R_{ct} at 150 mg L^{-1} AB1 is ten times larger than VMS. Overall, CV and EIS analyses confirm that AB1 plays a suppressive role during copper electrodeposition.

3.1.3 Chronopotentiometry analysis. It is important to study the interaction among SPS (accelerator), PEG (suppressor), and AB1 (leveler), as it provides a deeper understanding of the filling mechanism in microvias.[45] Traditionally, galvanostatic measurement is applied to study possible interactions by changing the injection order of the three types of additives. Fig. 1c (up) shows the change in cathodic potential after injecting additives into the testing solution. The cathodic potential declines (230 mV) after adding PEG, indicating its suppressing effect. Upon introducing AB1, the cathodic potential further decreases (31 mV), suggesting synergistic cooperation between AB1 and PEG. However, after the addition of SPS, the cathodic potential shifts positively (42 mV), indicating the accelerating effect of SPS in the presence of AB1 and PEG. By altering the additive injection order, SPS can cause a positive shift of 205 mV in cathodic potential after PEG injection, suggesting that SPS can replace PEG to accelerate copper electrodeposition, consistent with previous work.[46] Following AB1 injection, the cathodic potential changes negatively (−191 mV). Additionally, the potential reaches the same level regardless of injection order, further verifying the competition between SPS and AB1. Fig. 1c (down) provides a schematic diagram illustrating the relationships among SPS, PEG, and AB1, showing cooperation between AB1 and PEG, but competition between AB1 and SPS.

3.2 Effect mechanism study

3.2.1 Adsorption analysis. The influencing mechanism was studied using DFT calculations and *in situ* infrared spectroscopy.[47–49] The AB1 formula and its optimized structure are shown in Fig. S2 (ESI†). Fig. 2a shows the local electron affinity (LEA) distribution, with the blue color indicating high electrophilicity and red representing nucleophilicity. The $-SO_3$ area (high electrophilicity) in AB1 is conducive to forming interactions with the copper substrate due to vacant d orbitals. The adsorption equilibrium configuration of AB1 on the copper substrate is shown in Fig. 2b and Fig. S3 (ESI†), with color representations detailed in the ESI.† The quasi-plane structure of AB1 adsorbs nearly parallel to the copper substrate, with the $-SO_3$ functional group connected to the benzene ring of AB1 preferentially adsorbing on the substrate. Fig. 2c shows the 2D charge density difference of the adsorption configuration after equilibrium. The blue and red colors represent charge depletion and accumulation, demonstrating charge transfer between copper and AB1. It is believed that AB1 adsorbs on the cathode surface to reduce active sites and increase cathodic polarization.

To verify the calculated results, *in situ* infrared spectroscopy was conducted to observe the AB1 adsorption process during electrodeposition,[30] as shown in Fig. 2d. The peaks and their attributions are summarized in Table S1 (ESI†). The characteristic peaks ($-SO_3$ functional group) at 1080 cm^{-1} and 1220 cm^{-1}

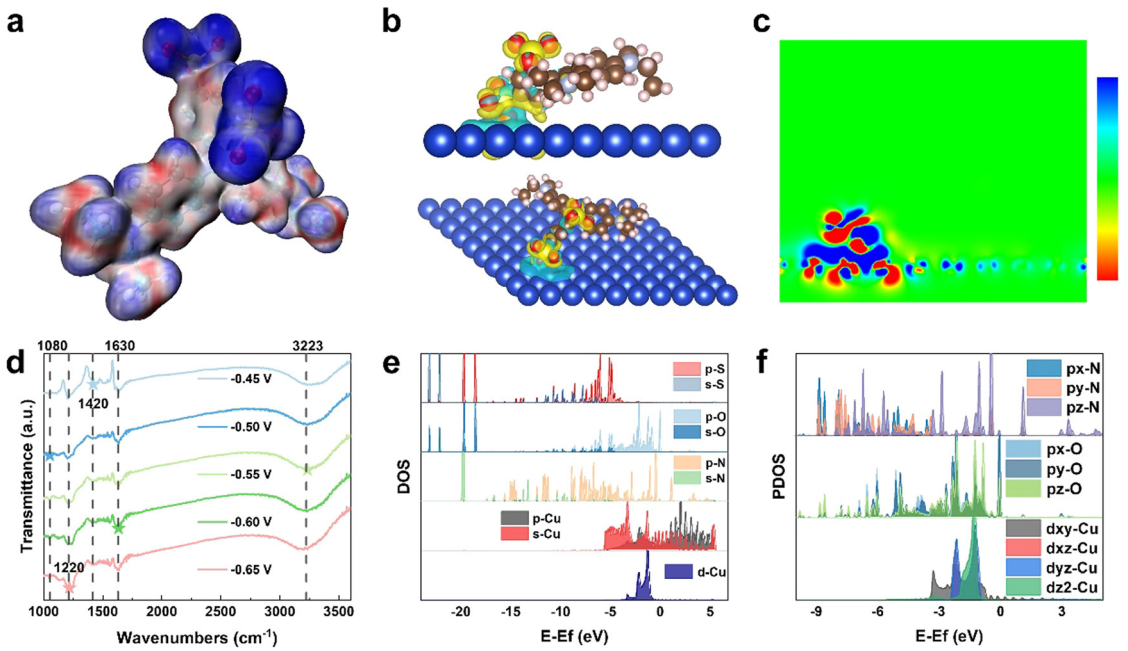


Fig. 2 (a) Local electron affinity (LEA) mapped the distribution of AB1. (b) Adsorption equilibrium of AB1 on the copper substrate (bottom) and side view of the adsorption equilibrium configuration (top). (c) 2D Bader charge density difference after AB1 adsorbs on the copper substrate. (d) *In situ* infrared spectroscopy of AB1 during electroplating. (e) DOS of s and p orbitals of S, O, and N from AB1 and s, p, and d orbitals of Cu from the copper substrate. (f) PDOS of the p orbitals of N and O from AB1 and the d orbitals of Cu from the copper substrate.

confirm the adsorption of the $-SO_3$ functional group of AB1 on the copper substrate. The stretching vibrations of C–N (1630 cm^{-1}), C–C (1420 cm^{-1}), and C–H (3223 cm^{-1}) of the aromatic ring indicate that AB1 adsorbs parallel to the copper substrate, consistent with the calculated adsorption equilibrium.

3.2.2 DOS analysis. Density of states (DOS) analysis provides insight into the interaction at the orbital level.[32] Fig. 2e shows the clear orbital overlap among the d orbitals of Cu, the s orbitals of N, and the p orbitals of O, indicating orbital interactions among the copper substrate, N atoms, and O atoms. In contrast, there is no orbital overlap between S and Cu, demonstrating that S atoms are not directly adsorbed on the copper substrate, consistent with the calculated adsorption configuration. To understand the specific orbital interactions in more detail, the PDOS (projected density of state) was calculated, as shown in Fig. 2f. Firstly, the pz orbitals of N atoms exhibit the largest overlap with the dyz and dz2 orbitals of Cu, while a strong hybridization occurs between py and pz orbitals of O atoms and dyz and dz2 orbitals of Cu. DOS analysis confirms significant d and p orbital hybridization among O, N, and Cu, further verifying the adsorption configuration and providing a deeper understanding of the adsorption mechanism. In summary, AB1 can adsorb on the copper cathode to reduce active sites and act as a barrier to suppress the reduction of free Cu^{2+}.

3.3 Effect of AB1 on filling performance

3.3.1 Electroplating performance. The optimization of the electroplating process is shown in Fig. 3, with the used devices illustrated in Fig. 3c. Without AB1, the growth rate at the opening area is higher than that at the bottom and side walls, leading to void defects in microvias with both diameters (125 μm and 150 μm). At an AB1 concentration of 10 mg L^{-1}, the growth rate at the opening area decreases due to AB1 adsorption, which suppresses copper growth. However, this relatively low concentration is not conducive to establishing an effective concentration gradient, resulting in defects forming in the 125 μm microvias and concave growth in the 150 μm microvias. With 20 mg L^{-1} AB1, an effective concentration gradient is formed, allowing copper to grow from bottom to top without defects in either microvias. Besides, the compact and flat copper morphology from the top view in Fig. S4 (ESI†) indicates that AB1 improves copper quality compared to layers obtained without it. In contrast, the commonly used leveler Janus Green B (Fig. S5, ESI†) produces a copper layer that contains gas holes, despite yielding a flat and defect-free surface, demonstrating that AB1 is superior in enhancing copper quality. The influence of morphology will be discussed in Section 3.4.1. When the AB1 concentration increases further to 30 mg L^{-1}, the excessive concentration weakens the concentration gradient, causing the bottom-up growth mode to shift to a conformal growth mode in the 150 μm microvias. Overall, 20 mg L^{-1} AB1 is determined to be optimum.

The stability of AB1 was later evaluated, as shown in Fig. S6 and S7 (ESI†). Cyclic voltammetry indicates that there is no cathodic reduction reaction peak in the potential range of −0.75 V to +0.2 V *vs.* SSE, illustrating that no decomposition occurs for AB1 during electroplating. After introducing different coulombs (1 A h, 3 A h, 5 A h) into the solution with the optimized AB1 concentration (20 mg L^{-1}), electroplating experiments were carried out and the cross-sectional figure of the copper layer is shown in Fig. S7 (ESI†). A defect-free copper

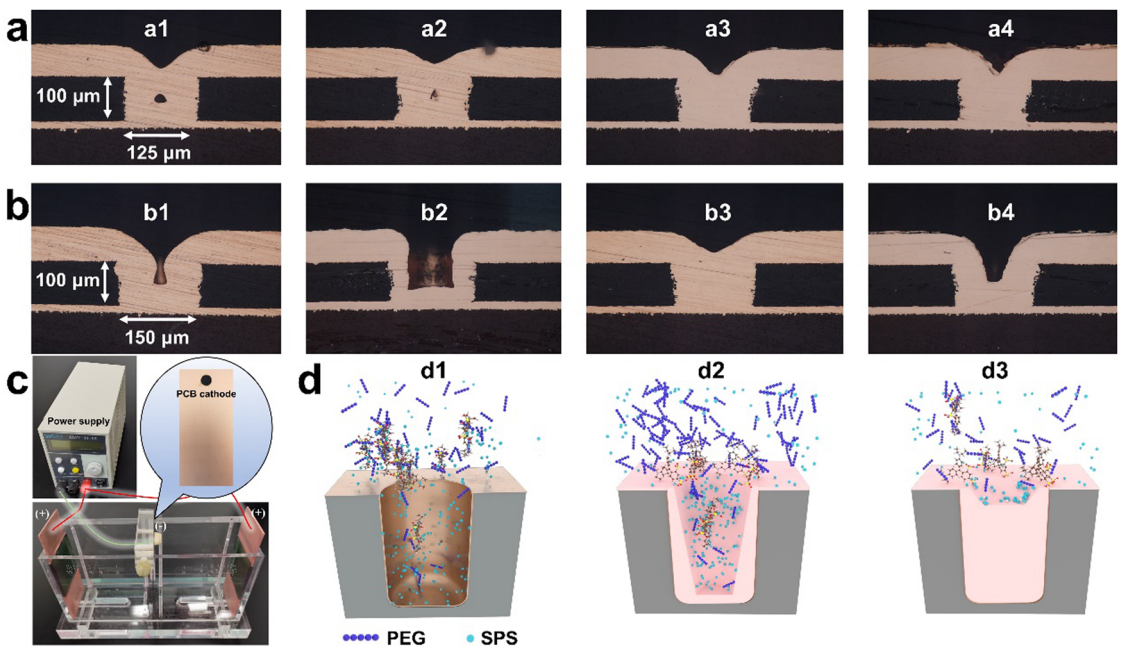


Fig. 3 Filling performance of Cu electroplating for 3 h with different AB1 concentrations in microvias with diameters of (a) 125 μm, (b) 150 μm. (a1) and (b1) (0 mg L^{-1}), (a2) and (b2) (10 mg L^{-1}), (a3) and (b3) (20 mg L^{-1}), and (a4) and (b4) (30 mg L^{-1}). (c) Electroplating devices. (d) Schematic diagram of the filling process for AB1–SPS–PEG system.

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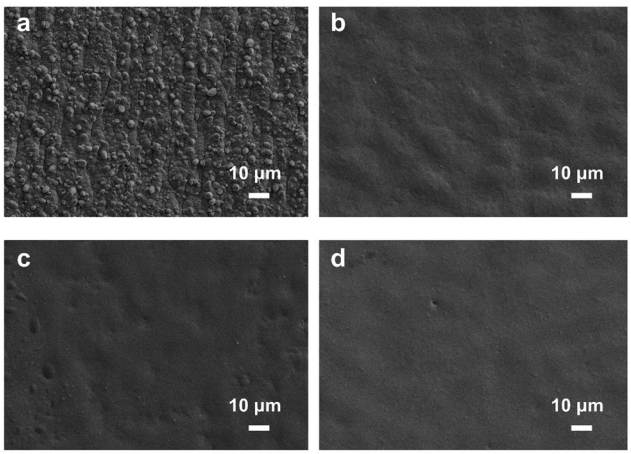


Fig. 4 Morphology of the copper layer electrodeposited with different AB1 concentrations. (a) 0 mg L^{-1}, (b) 10 mg L^{-1}, (c) 20 mg L^{-1}, and (d) 30 mg L^{-1}.

layer was obtained, further verifying that AB1 maintains good stability during electroplating.

3.3.2 Filling mechanism. The diffusion property of AB1 was studied using molecular dynamics (MD) simulation, as shown in Fig. S8 (ESI†). Based on the Einstein equation (eqn (S2), ESI†), the diffusion coefficient for AB1 (D_{AB1}) is 2×10^{-10} m^2 s^{-1}, which is close to the diffusion coefficient for PEG (D_{PEG}, 1×10^{-10} m^2 s^{-1})[50] and one order smaller than that for SPS (D_{SPS}, 5×10^{-9} m^2 s^{-1}).[51] This illustrates that the leveler AB1 and suppressor PEG have slower diffusion transport compared to the accelerator SPS. Coupled with the interactions among the three additives discussed in Section 3.1.3, namely the competition between AB1 and SPS, as well as cooperation between AB1 and PEG, gradient competitive adsorption superfilling was proposed to illustrate the AB1–SPS–PEG system, with a schematic diagram shown in Fig. 3d. At the initial stage (3d1), different additives are distributed uniformly within the microvias. As the filling process progresses (3d2), a concentration gradient for the additives forms due to their consumption and local diffusion transport limitations within the microvias. However, the concentration ratio of SPS to PEG and AB1 at the bottom increases compared to that at the opening area due to the differing diffusion properties among the three additives; specifically, D_{SPS} is significantly larger than D_{AB1} and D_{PEG}. As a result, the growth rate at the bottom is improved by the increased accelerating effect from the relatively amount of SPS, while growth at the opening area is effectively suppressed since AB1's mass transport is not limited due to convection, leading to copper growth from the bottom to the top. At the final stage (3d3), the three additives redistribute uniformly at the opening area, resulting in the same growth rate across different areas. No momentum plating phenomena occur in the AB1–SPS–PEG system.[52]

3.4 Effect of AB1 on the copper layer

3.4.1 Effect of AB1 on morphology. Fig. 4 and 5a show the surface morphology of the copper layer observed by SEM and Olympus conformal laser microscope.[53] Small bump particles are present on the uneven copper surface electrodeposited without AB1. After introducing 10 mg L^{-1} AB1, the surface flatness improves greatly, and the copper layer becomes

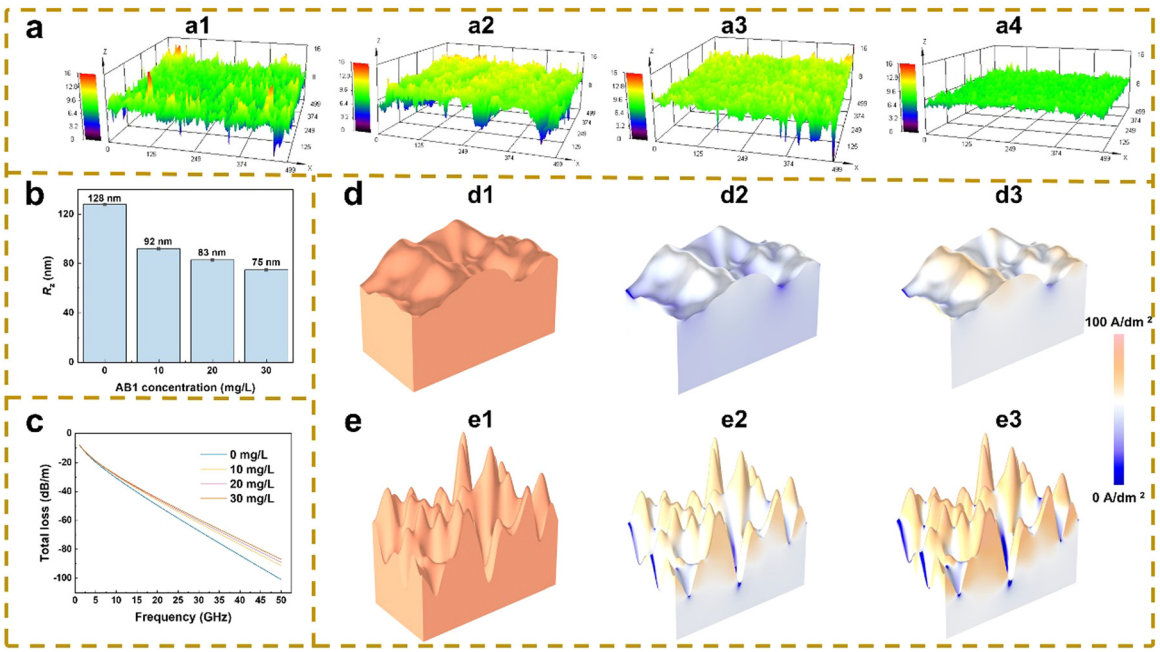


Fig. 5 (a) Surface roughness observed from the Olympus conformal laser microscope. (b) Histogram of surface roughness (R_z) influenced by AB1 concentration. (c) Relationship between signal transmission loss and AB1 concentration at different frequencies. Simulated current density distribution in the copper layer with 20 mg L^{-1} AB1 (d) and without AB1 (e) when applied frequencies are 10 GHz (d2) and (c2) and 100 GHz (d3) and (c3). The simulation model shown in (d1) and (c1) was established based on R_z.

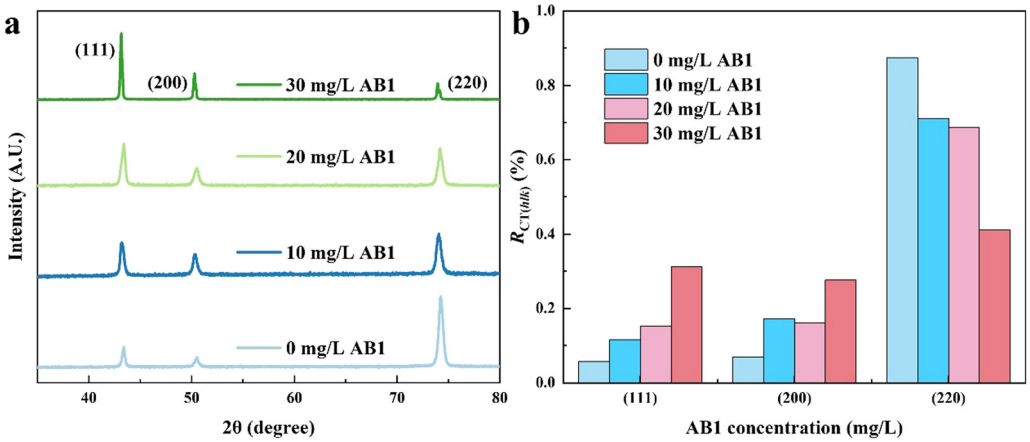


Fig. 6 (a) XRD pattern of copper layer electrodeposited with different AB1 concentrations. (b) Histogram of calculated $R_{CT(111)}$, $R_{CT(200)}$, and $R_{CT(220)}$.

compact, although large bump morphology remains. As the AB1 concentration continues to rise, the flatness of the copper layer further improves. Using the Olympus conformal laser microscope, the surface roughness (R_z) was calculated, as shown in Fig. 5b. R_z is 128 nm for the copper layer obtained without AB1 and decreases to 92 nm with 10 mg L^{-1} AB1. Furthermore, R_z values obtained from 20 mg L^{-1} and 30 mg L^{-1} are 83 nm and 75 nm, respectively. The above analysis indicates that AB1 has a micro-scale leveling effect.

3.4.2 Effect of AB1 on signal loss. The influence of AB1 on signal loss during high-frequency transmission was studied using finite element simulation.[54] Fig. 5c shows that signal loss is influenced by surface roughness and applied frequency. Total loss increases with rising frequency and R_z. Additionally, the total loss with frequency for the copper layer obtained with JGB is also calculated (Fig. S5d, ESI†). In comparison, the signal loss obtained with AB1 is smaller than with JGB at the same frequency due to a lower R_z for AB1. The skin effect for the optimized copper layer with 20 mg L^{-1} AB1 and without AB1 is shown in Fig. 5d and e. At a relatively low frequency (10 GHz), current density is distributed uniformly in the copper layer obtained from 20 mg L^{-1}. Even at high frequency (100 GHz), the current density remains relatively well distributed. However, for the copper layer obtained without AB1, the current density mainly concentrates near the surface area at 10 GHz. Simulation results indicate that AB1 helps mitigate the influence of the skin effect during high-frequency signal transmission by improving surface flatness.

Table 2 Calculated grain size of the copper layer obtained with different AB1 concentrations from XRD patterns according to Scherrer's equation

AB1 concentration (mg L^{-1})	Grain size (nm)			
	(111)	(200)	(220)	Mean grain size (nm)
0	29.0	25.3	24.1	26.1
10	22.7	21.6	22.5	22.3
20	19.6	17.1	20.6	19.1
30	16.7	14.2	17.1	16.9

3.4.3 Effect of AB1 on phase structure. Fig. 6a shows the XRD of the copper layer with peaks at 43.3°, 50.4°, and 74.1° corresponding to Cu(111), Cu(200), and Cu(220), respectively. The texture coefficient $R_{CT(hkl)}$ was calculated from the XRD to show the influence of AB1 on crystal orientation as follows:[55]

$$R_{CT(hkl)} = \frac{I_{(hkl)}/I_{0(hkl)}}{\sum\left(I_{(hkl)}/I_{0(hkl)}\right)} \quad (3)$$

where $I_{0(hkl)}$, $I_{(hkl)}$ represent the normalized intensity of the standard copper sample and the copper layer obtained with different AB1 concentrations, as shown in Fig. 6b. $R_{CT(111)}$ and $R_{CT(200)}$ increase with rising AB1 concentrations, while $R_{CT(220)}$ continuously decreases, demonstrating that AB1 is beneficial for forming of Cu(111) and Cu(200). Furthermore, Scherrer's formula was used to calculate the influence of AB1 on the grain size of the copper layer, as shown in Table 2. The grain size of copper is 26.1 nm without AB1, while it decreases continuously to 16.9 nm with increased AB1 concentration, indicating that AB1 promotes grain refinement.

4. Conclusion

A novel leveler, Acid Blue 1, was used to achieve void-free copper filling in blind microvias. Electrochemical tests (CV and EIS) confirm the suppressive effect on the reduction of Cu^{2+}. During copper electroplating, there is a synergistic suppression between AB1 and PEG, but competition between AB1 and SPS. DFT calculations and *in situ* infrared spectroscopy provide a deeper understanding of AB1 adsorption configuration on the copper substrate during electrodeposition, indicating that AB1 can reduce active sites and act as a barrier adsorption layer to increase cathodic polarization. After electroplating optimization, microvias with diameters of 125 μm and 150 μm can be filled with 20 mg L^{-1} AB1 at 1.5 A dm^{-2}; the good stability of AB1 indicates its industrial potential. A gradient competitive adsorption superfilling mechanism was proposed to explain the filling process for the AB1–SPS–PEG system. The diffusion differences among the three additives

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and the competition between AB1 and SPS are key to achieving bottom-up growth in microvias. AB1 improves the copper layer quality, enhances surface flatness, refines the grains, and reduces signal losses during high-frequency transmission. Additionally, AB1 suppresses the formation of Cu(220) while promoting the formation of Cu(111) and Cu(200).

Data availability

All data are provided in full in the results section of this paper and the ESI.†

Conflicts of interest

There are no conflicts to declare.

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